EXPERIMENT 26 : Implementation of Best First Search Algorithm Using Prolog

Aim:

To write and execute a Prolog program to implement the Best First Search algorithm.

Software Required:

- SWI-Prolog
- Windows Operating System

Theory:

Best First Search is an informed search algorithm used in Artificial Intelligence. It selects the most promising node based on its heuristic value. The node with the lowest heuristic value is selected first and expanded until the goal node is reached.

Program:

Save the program as exp22.pl in C:/AI Prolog/.

```
% Best First Search Algorithm

% Graph representation
edge(a,b).
edge(a,c).
edge(b,d).
edge(b,e).
edge(c,f).
edge(c,g).
edge(d,h).
edge(e,h).
edge(f,h).
edge(g,h).

% Heuristic values
h(a,7).
h(b,6).
```

```
h(c,4).
h(d,5).
h(e,3).
h(f,3).
h(g,1).
h(h,0).

% Best First Search
bestfirst(Start,Goal,Path) :-
    search([[Start]],Goal,RevPath),
    reverse(RevPath,Path).

search([[Goal|Path]|_],Goal,[Goal|Path]).

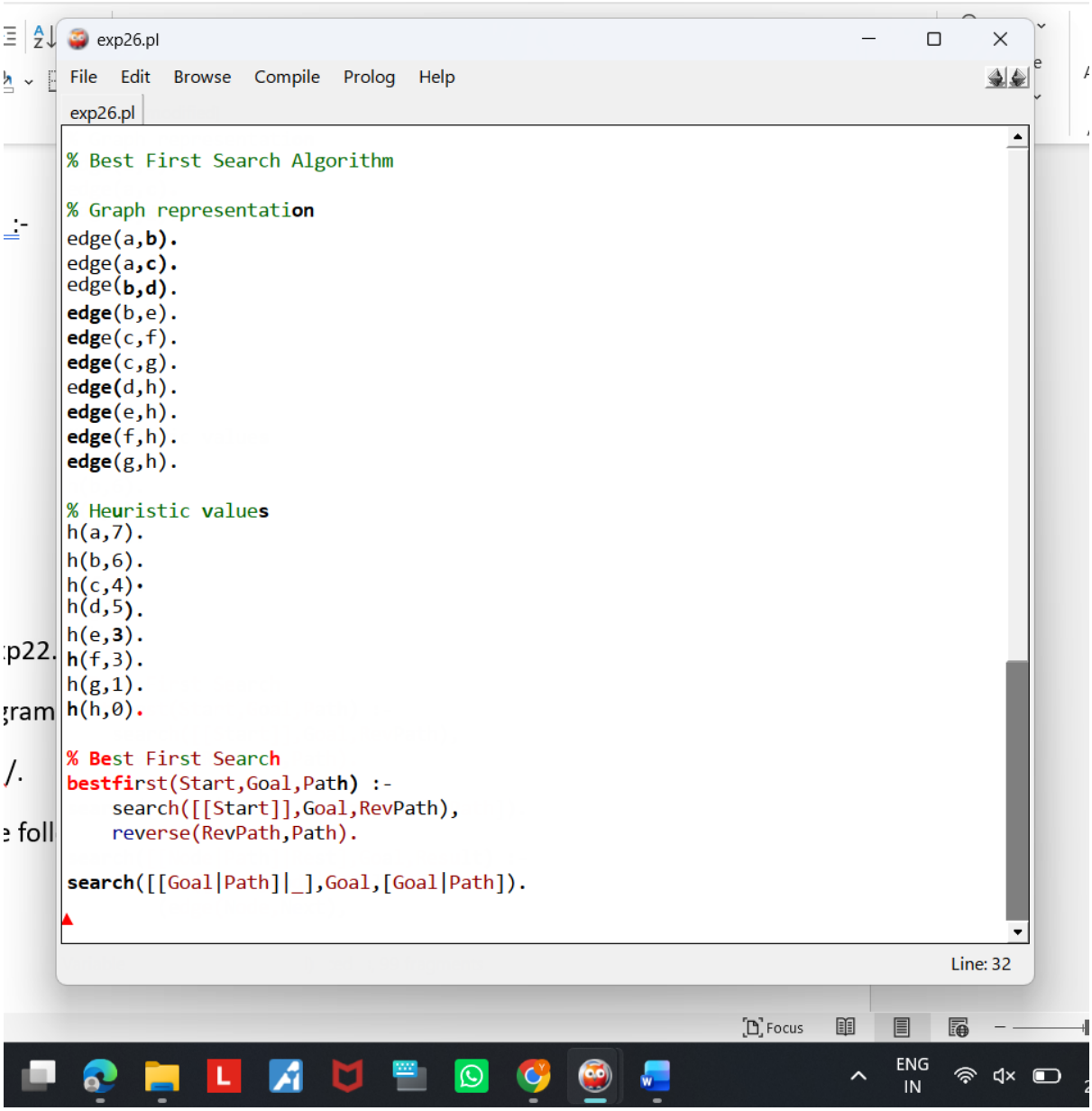
search([[Node|Path]|Rest],Goal,Result) :-
    findall([Next,Node|Path],
        (edge(Node,Next),
         \+ member(Next,[Node|Path])),
        Children),
    append(Rest,Children,NewOpen),
    sort_paths(NewOpen,Sorted),
    search(Sorted,Goal,Result).

% Sort according to heuristic value
sort_paths(Paths,Sorted) :-
    predsort(compare_paths,Paths,Sorted).

compare_paths(Order,[A|_],[B|_]) :-
    h(A,HA),
```

h(B,HB),

compare(Order,HA,HB).



Procedure:

1. Open SWI-Prolog.
2. Create a new file named exp22.pl.
3. Type the above Prolog program.
4. Save the file in C:/AI Prolog/.
5. Load the program using the following command:

?- consult('c:/ai prolog/exp22.pl').

6. Execute the following query:

?- bestfirst(a,h,Path).

Output:

```
ERROR: Unknown procedure: dit/1 (DWIM could not correct goal)
64 ?- consult('c:/ai prolog/exp26.pl').
true.

65 ?- bestfirst(a,h,Path).
Path = [a, c, g, h] .

66 ?- bestfirst(a,g,Path).
Path = [a, c, g]
```

Result:

Thus, the Prolog program to implement the Best First Search algorithm was successfully written and executed using SWI-Prolog.